Lecture 05 - Introduction

Lecture 05 - Distributions and the Normal Model - Introduction

Purpose of the Lesson

This lecture introduces continuous probability distributions and the normal model. Lecture 4 introduced random variables in general. Lecture 5 focuses on the distributions that describe continuous quantities such as returns, waiting times, demand, delivery time, and other measured business or financial variables.

The main conceptual shift is that continuous variables are not handled by assigning probability to one exact value. Instead, probabilities are assigned to intervals. A density curve describes how probability is spread across the possible values, and probability is represented by area under that curve.

The normal model receives special attention because it is one of the central benchmarks in statistics. It is simple, widely used, and mathematically convenient. At the same time, it can be misleading when the data are skewed, bounded, multi-modal, or heavy-tailed. The lecture therefore treats the normal model as a useful tool that must be checked, not as an automatic truth.

Python appears as a support tool for computing probabilities, plotting empirical distributions, fitting a normal benchmark, and comparing model probabilities with observed frequencies. The goal is not to memorize code. The goal is to understand what the code is computing and how to interpret it carefully.

Learning Objectives

This lecture aims to:

1. Explain the difference between discrete and continuous probability distributions.
2. Introduce probability density functions and the idea that continuous probability is area.
3. Describe the uniform, exponential, and normal distributions.
4. Explain how the mean and standard deviation shape a normal distribution.
5. Use the cumulative distribution function to compute normal probabilities.
6. Introduce standardization and z-scores.
7. Fit a normal model to empirical return data.
8. Compare empirical probabilities with probabilities implied by a fitted normal model.
9. Explain why the normal model can be useful near the center of a distribution but misleading in the tails.

Learning Outcomes

By the end of this session, students should be able to:

1. Explain why (P(X=x)=0) for an exact value in a continuous model.
2. Interpret (P(a<X<b)) as an area under a density curve.

3. Identify when a uniform, exponential, or normal distribution might be a plausible model.
4. Compute and interpret simple probabilities under a uniform distribution.
5. Compute z-scores and interpret them as distances in standard deviations.
6. Use a normal CDF to compute probabilities below a threshold or inside an interval.
7. Distinguish an empirical distribution from a fitted probability model.
8. Fit a normal model using a sample mean and sample standard deviation.
9. Compare empirical tail frequencies with normal-model tail probabilities.
10. Write careful model-based interpretations.

Lesson Roadmap

The lecture begins by contrasting discrete and continuous random variables. In a discrete model, probabilities can be assigned to exact values. In a continuous model, probabilities are assigned to intervals. This leads to the central concept of a probability density function.

The first examples are the uniform distribution and the exponential distribution. The uniform distribution shows the simplest possible density: a flat rectangle over an interval. The exponential distribution introduces a right-skewed density used for waiting times.

The lecture then turns to the normal distribution. We discuss its bell shape, its symmetry, and the roles of the mean and standard deviation. We then show how probabilities under the normal curve are computed with the cumulative distribution function.

After that, the lecture introduces standardization. A z-score transforms a value into the number of standard deviations it lies above or below the mean. This provides a common scale for comparing values from different normal distributions.

The final part moves from theory to data. We fit a normal distribution to Netflix daily log returns and compare the fitted model with empirical frequencies. The comparison shows a key practical lesson: the normal model may be useful as a benchmark, but financial returns can have heavier tails than the normal distribution.

Central Idea

The central idea of the lecture is:

For continuous variables, probabilities are areas under a model curve, and the model must be checked against the data.

For example, suppose a normal model says that the probability of a large negative return is 0.135%. That number is not a law of nature. It is the probability implied by the model. If the observed data show large negative returns much more often, the model is underestimating tail risk.

This distinction is essential for finance. Normal approximations are common, but risk often lives in the tails. A model that looks reasonable in the center can still be dangerous if it understates rare but severe losses.

How to Work Through the Lesson

Begin by focusing on the language of continuous probability. Make sure you can explain why the probability of an interval is meaningful but the probability of one exact point is zero in a continuous model.

Then study the shapes of the three distributions. The uniform distribution is flat, the exponential distribution is right-skewed, and the normal distribution is symmetric. These shapes are not decoration. They determine which probabilities are large and which are small.

When working with the normal model, keep the roles of (\mu) and (\sigma) separate. The mean locates the center. The standard deviation controls spread. Then practice converting values to z-scores and using the CDF logic (P(a<X<b)=F(b)-F(a)).

Finally, pay attention to the empirical comparison. A fitted normal curve is a model. It should be compared with histograms, observed frequencies, and especially tail events. In finance, this checking step is not optional.

What to Pay Attention To

Three habits matter in this lecture.

First, separate density from probability. The height of a density curve is not itself a probability. Probability is area over an interval.

Second, separate empirical facts from model statements. A histogram summarizes observed data. A normal curve is a theoretical approximation fitted to those data.

Third, use cautious language. Say "under the fitted normal model" when reporting a model probability. Say "in the observed sample" when reporting an empirical frequency. This language prevents a useful approximation from being mistaken for reality.

End-of-Lesson Check

By the end of the lecture, you should be comfortable answering questions such as:

1. Why do continuous random variables assign probability to intervals rather than exact values?
2. What conditions must a probability density function satisfy?
3. Why is probability represented by area under a density curve?
4. What is the density of a uniform distribution on ([a,b])?
5. Why is the exponential distribution useful for waiting times?
6. What do (\mu) and (\sigma) control in a normal distribution?
7. How do we compute (P(a<X<b)) using a CDF?
8. What is a z-score?
9. What is the difference between empirical and fitted distributions?
10. Why can a normal model be useful but still underestimate tail risk?

If these questions are clear, you are ready to move to sampling distributions and confidence intervals in Lecture 7.